

Student Utility Program – Lab Report

1. Aim

To develop a Java console-based Student Utility Program using classes, objects, methods, a switch statement, and a do-while loop to enter student details, calculate total marks and percentage, display the result, and exit the program.

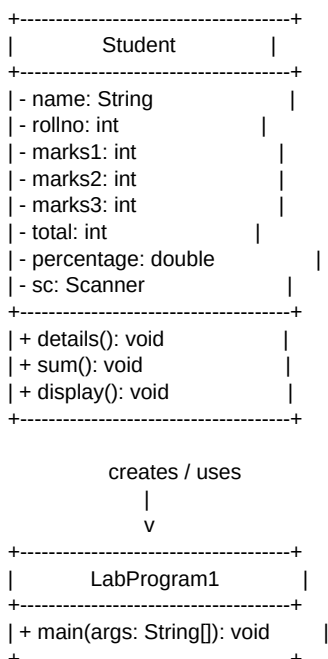
2. Description

The program belongs to the package **observationtasks** and contains two classes: **Student** and **LabProgram1**. The Student class stores the student's name, roll number, three subject marks, total, and percentage. The details() method accepts input, sum() calculates the total and percentage, and display() prints the complete result. The LabProgram1 class contains the main() method and presents a menu repeatedly until the user selects option 4.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Execution Tool: Java Development Kit (JDK)
- Input Method: java.util.Scanner
- Concepts Used: Classes, objects, methods, switch-case, do-while loop, and arithmetic operators

4. UML Class Diagram



5. Algorithm

1. Start the program.
2. Create a Scanner object and a Student object.
3. Display the Student Utility Program menu.

4. Read the user's choice.
5. If the choice is 1, call details() and accept the student's name, roll number, and three marks.
6. If the choice is 2, call sum(), calculate the total, and calculate the percentage.
7. If the choice is 3, call display() and show the student information and result.
8. If the choice is 4, display the exit message and terminate the loop.
9. For any other choice, display an invalid-choice message.
10. Repeat the menu while the choice is not 4.
11. Close the Scanner and stop the program.


```
        case 2:
            st.sum();
            break;

        case 3:
            st.display();
            break;

        case 4:
            System.out.println("Program Exited.");
            break;

        default:
            System.out.println("Invalid Choice!");
    }

} while (choice != 4);

sc.close();
}
```

7. Sample Output

The following is an example execution of the program using sample student details.

```
--- Student Utility Program ---
1. Enter Student Details
2. Calculate Total and Percentage
3. Show Result
4. Exit
Enter your choice: 1
```

```
--- Student Details ---
Enter Student Name: Srikar
Enter Roll Number: 101
Enter Marks 1: 85
Enter Marks 2: 90
Enter Marks 3: 80
```

```
--- Student Utility Program ---
1. Enter Student Details
2. Calculate Total and Percentage
3. Show Result
4. Exit
Enter your choice: 2
Total Marks = 255
Percentage = 85.0
```

```
--- Student Utility Program ---
1. Enter Student Details
2. Calculate Total and Percentage
3. Show Result
4. Exit
Enter your choice: 3
```

```
--- Student Result ---
Name      : Srikar
Roll Number : 101
Marks 1    : 85
Marks 2    : 90
Marks 3    : 80
Total      : 255
Percentage : 85.0
```

```
--- Student Utility Program ---
1. Enter Student Details
2. Calculate Total and Percentage
3. Show Result
4. Exit
Enter your choice: 4
Program Exited.
```

8. Result

The Student Utility Program was successfully designed in Java. It accepts student details, calculates the total marks and percentage for three subjects, and displays the student result through a menu-driven interface. The program demonstrates the use of object-oriented programming basics, methods, conditional selection, and looping.